Arush Bodla

LinkedIn | abodla@umd.edu | Portfolio | GitHub

Education

University of Maryland – Banneker/Key Scholar College Park, MD

B.S. in Computer Science — ACES Cybersecurity Honors *May 2030*

Relevant Coursework: Data Structures, Algorithms, Object-Oriented Programming, Discrete Structures, Computer Systems

Experience

Digital Forensics Intern June 2026 – Aug. 2026

ManTech — DFEND Program *Herndon, VA*

- Cracked "presto," a stripped, self-modifying binary reputed to take weeks, in 3 days: reversed 33 XOR-encrypted check-functions (RC4, CRC32) and recovered all 666 key segments via a 20-minute brute force.
- Emulated check-functions with GDB's Python API, then Unicorn CPU emulation; patched out anti-debugging routines and edited process memory directly.

Linux Captain Aug. 2026 – Present

UMD Collegiate Cyber Defense Competition (CCDC) *College Park, MD*

- Built the team's DevOps framework from scratch: Docker-based tests run every script as root and unprivileged across bash, dash, ksh, posh and busybox; GitHub Actions CI on every push.
- Wrote **umdinate**, a parallel SSH deployer with per-service YAML playbooks and a safety gate blocking untested scripts from root; caught by reading source that the underlying tool reported failed sudo as success, and fixed it.
- Built a 4-script SSH hardening suite: `sshd -t`/`-T`-verified config, auto-revert unless a new login is confirmed, PAM backdoor checks.

Co-Founder & Tournament Director Sept. 2023 – June 2026

ACL Competitive Programming Club *Leesburg, VA*

- Co-founded and ran ACL-IT, a beginner programming tournament (2 editions); authored 12 problems.
- Architected the AWS judging stack (DOMjudge on 3 EC2 instances, judgehosts isolated in a VPC, custom AMIs for recovery), sustaining 500+ submissions in a 3-hour contest.
- Built ACL-OJ, an offline-first LAN judge that keeps contests AI-free; cut verdict latency about 80%.

Projects

Ghost Racer (67 Racer) | *PyTorch, ROS 2, MQTT, ESP32, AWS DeepRacer* GitHub | Devpost

- Built a behavioral-cloning car: hand gestures steer a physical AWS DeepRacer, then a CNN trained on that driving pilots it autonomously. **2nd overall, AcademiesHacks 2026.**
- Owned the hardware bring-up: wired batteries and sensors, flashed Ubuntu, and with teammates reverse-engineered Amazon's discontinued DeepRacer API for direct ROS 2 control.

RegiMon | *Rust, Tauri, React, Win32 API, GitHub Actions* GitHub

- Co-built an event-driven Windows registry monitor that catches in-memory malware persistence: under 50 ms detection, under 1% CPU across 50 keys, one-click rollback.
- Caught and reverted LoveBug persistence Windows Defender missed, across 25 MITRE ATT&CK-scoped keys.

Nexus | *LangGraph, RAG, Next.js, Supabase, FastAPI* GitHub | Devpost

- Architected a 7-node multi-agent LangGraph pipeline that turns a student's learning gap into a playable game in minutes. **Best Education, PatriotHacks 2026.**
- Kept AI output coherent with ten genre templates and a RAG index over engine docs; owned Supabase schema and RBAC.

Awards & Honors

- **Competitive Programming:** IIOT Team USA; 1st UPenn Philadelphia Classic; USACO Gold; Codeforces 1530
- **Cybersecurity:** CyberPatriot 13th of 4,787 nationally; NCL 4th nationally (HS); 1st GTEdge AI CTF
- **Hackathons (2026):** 1st IndyHAX; 2nd AcademiesHacks; 3rd Fetch.ai track MHacks; Best Education PatriotHacks

Technical Skills

Languages: C++, Python, Java, Rust, C#, TypeScript, Bash/POSIX sh
Systems & Hardware: Linux, Docker, AWS, ROS 2, ESP32, Win32 API, GitHub Actions, Tauri
ML & Low-Level: PyTorch, LangGraph, RAG, Ghidra, gdb, Unicorn, Frida